HERIOT WATT UNIVERSITY

DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

F78AA Actuarial and Financial Mathematics A

Semester 1 2015–16

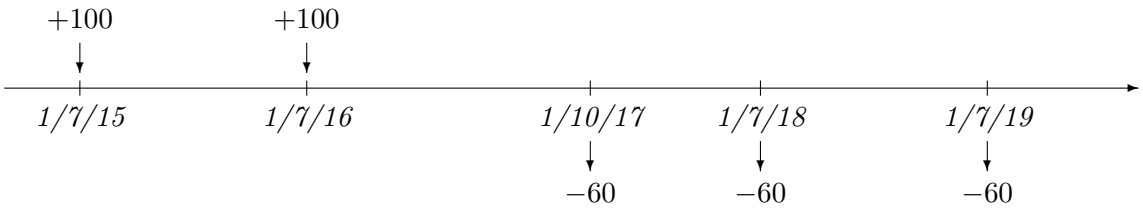
2 Hours

Question	Marks
1	7
2	5
3	6
4	10
5	9
6	7
7	16
Total Marks	60

Attempt ALL questions.

Approved electronic calculators may be used.

1. The effective rate of interest will be 4% per annum during 2015, 2% per half-year during 2016 and 2017, and 1% per quarter-year thereafter. A series of cashflows is given as follows:



(a) What is the present value on 1 January 2015 of these cashflows? [5 marks]

(b) What is the accumulation on 1 July 2020 of these cashflows? [2 marks]

(Minor discrepancies caused by different roundings will not be penalised.)

[Total 7 marks]

2. Let i be the effective rate of interest per time unit.

(a) Write down a precise definition in words of $\ddot{s}^{i}_{\overline{n}|}$. [1 mark]

(b) Show from first principles that $\ddot{s}^{i}_{\overline{n}|} = \dfrac{(1+i)^n - 1}{d}$. [4 marks]

[Total 5 marks]

3. An annuity-certain is payable annually in arrear for 15 years. The first payment is £10,000, and subsequent payments increase at rate 0.02 (that is, by 2% compound) each year. During the first 10 years the effective rate of interest is 4% per year, thereafter it is 6% per year. What is the accumulation of this annuity at the end of the payment period?

(Minor discrepancies caused by different roundings will not be penalised.)

[6 marks]

continued ...

4. (a) You borrow £200,000 to buy a house. The loan is to be repaid over 30 years by level annual repayments, payable in arrear. The interest rate is 7% per annum effective.

   (i) What is the level annual repayment?

   (ii) What is the capital outstanding just after the 10th repayment has been made?

   (iii) What is the interest content of the 11th repayment?

   [5 marks]

   (b) Just after the 10th repayment has been made, you inherit some money. You ask the lender if you can make repayments of £25,000 for the next three years, and reduced repayments thereafter. The lender agrees, provided that, at the end of the three years, you switch from making annual repayments to making monthly repayments. What is the monthly repayment that you will pay after three years? [5 marks]
   (Minor discrepancies caused by different roundings will not be penalised.)

   [Total 10 marks]

5. A company issues a bond with term 25 years, to be redeemed at a premium of 2%, with a coupon of 8% per annum payable half-yearly in arrear.

   (a) Some of the bond is purchased at outset by an investor who pays income tax at 20% and capital gains tax at 20%. The investor pays a price P per £100 nominal, such that their net redemption yield is 7% per annum effective. What is P? [6 marks]

   (b) Immediately after payment of the 15th coupon, the investor sells the bond for exactly the same price P per £100 nominal as they paid for it. Explain why the net yield per annum that the investor achieves on the completed transaction is 6.90%. [3 marks]

   [Total 9 marks]

continued ...

6. An investment fund has assets worth £100,000 on 31 December 2013. It receives an inflow of £20,000 on 1 January 2015 and pays an outflow of £50,000 on 1 January 2017. On 31 December 2017 it has assets of £90,000.

   (a) Find the money-weighted rate of return per annum during the calendar years 2014–2017.

   [5 marks]

   (b) Suppose also that the fund has assets of £102,000 on 31 December 2014, £133,000 on 31 December 2015 and £129,000 on 31 December 2016. Find the time weighted rate of return per annum during the calendar years 2014–2017.

   [2 marks]

[Total 7 marks]

continued ...

7. An investor intends to borrow £5 million to fund a project. The borrowed money will be invested immediately. The project will take two years to become operational, and will deliver profits of £1.5 million at the end of each of the second to eighth years inclusive. The investor will pay interest of 7% per annum effective on any borrowings, and can earn interest of 6% per annum effective on invested money. If it is necessary to borrow money from the bank at any time, the investor will always repay such a bank loan as soon as possible. The investor can choose one of three methods finance to project.

Method A: The investor can issue a bond, with nominal amount £5 million, and term eight years, repayable at a premium of 5%. Coupons will be 8% per annum, payable yearly in arrear.

Method B: The investor can borrow the £5 million from a financial institution, and repay the loan by level annual repayments, payable in arrear.

Method C: The investor can borrow the £5 million from the bank, and use all profits earned to repay the loan as soon as possible.

(a) Show that the accumulated profit at the end of the eighth year under Method A is £3.376 million. [5 marks]

(b) Show that the accumulated profit at the end of the eighth year under Method B is £4.288 million. [5 marks]

(c) Let the discounted payback period under Method C be denoted k years. Show that:

$$k - 1 \geq \frac{\log 1.33274}{\log 1.07}$$

and hence that the accumulated profit at the end of the eighth year is £4.351 million. [6 marks]

[16 marks]

Total 60 marks

END OF PAPER